PENGWIN Task 1 v3.6.1 — Split-Aware Candidate RF

Algorithm Description · 2026-08-17

1. Submission status

`v3.6.1-split-aware-rf` is a separate upload candidate. The existing `v3.7` release is not modified. v3.6.1 retains the v3.5 Stage-A model, hybrid random-forest family router, largest-component ROI policy, anatomy-specific Stage-B experts, and all neural-network weights. It adds a trained split-candidate RF gate after Stage B. The prior image/model pair remains available until this candidate passes the Grand Challenge container test.

2. Method overview

The submission performs per-fragment instance segmentation of pelvic and femoral fractures with a serial two-stage STU-Net-B cascade:

1. **Stage A — anatomy.** A 5-class whole-volume model identifies sacrum, left/right hipbone and femur.
2. **Hybrid family routing.** A random forest selects pelvic or femur when confident; the organizers' acquisition rule is the uncertainty tiebreak.
3. **Stage B — fracture experts.** The routed anatomy selects a Sacrum, shared-Left/Right-Hip or Femur expert. It emits four ABBC channels and nine learned same-instance affinity channels at 1-, 3- and 9-voxel offsets.
4. **Safe base.** The exact v3.5 one-voxel average-linkage affinity decoder at `T=0.75` produces the default instance partition.
5. **Candidate proposal.** A parallel 1/3/9-voxel RAG-veto decoder is refined by full ABBC split/merge logic. It proposes local binary splits inside a v3.5 instance but never changes foreground support.
6. **Split-aware RF gate.** Five random-forest regressors estimate candidate changes in Merge, Split, Dice, Instance F1, and Precision from 44 inference-only geometry, CT, affinity, and ABBC context features. The learned conservative policy selects at most one candidate per source instance. Rejected sources remain exactly v3.5.
7. **Assembly.** Instances use the official ranges: sacrum 1-50, left hip 51-100, right hip 101-150 and femur 151-200.

3. Frozen model and RF contract

Component	Deployed value
Stage A	`Dataset539_PelvicFemurAnatomyV3`, `PengwinTrainerSTUNetBaseAnatomyV301`, fold 0
Family router	v3.3 hybrid RF, confidence margin 0.15

Component	Deployed value
Stage B dataset	`Dataset538_PelvicFemurBICMFragmentV5`
Stage B experts	`...V308SacrumExpertDeployedVal`, `...V308HipExpertDeployedVal`, `...V308FemurExpertDeployedVal`
Stage B fold/checkpoint	fold 0 / `checkpoint_best.pth`
Stage B output	13 channels: 4 ABBC + 9 affinity
Safe base	v3.5 one-voxel affinity agglomeration, `T=0.75`
Proposal	1/3/9 RAG veto, minimum 32 range pairs; 3 ABBC split passes, 400-voxel minimum piece
Candidate minimum	both binary pieces at least 1,000 mm3
RF inputs/outputs	44 inference-only features / five predicted metric deltas
RF policy	predicted Merge <= -0.55, Split <= 2.25, Dice >= -0.01, F1 >= -0.05, Precision >= -0.10

The Stage-A and Stage-B neural checkpoints and the family router are unchanged from v3.5. The model archive is new because it additionally contains five fitted candidate-outcome RF regressors and their fixed policy metadata.

4. Model archive layout

Grand Challenge extracts the archive directly under `/opt/ml/model`:

```
/opt/ml/model/

├── nnunet/results/

│   ├── Dataset539_PelvicFemurAnatomyV3/
```

```
│   │   └── PengwinTrainerSTUNetBaseAnatomyV301__.../fold_0/

│   └── Dataset538_PelvicFemurBICMFragmentV5/

│       ├── PengwinTrainerSTUNetBaseAffinityV308SacrumExpertDeployedVal__.../fold_0/

│       ├── PengwinTrainerSTUNetBaseAffinityV308HipExpertDeployedVal__.../fold_0/

│       └── PengwinTrainerSTUNetBaseAffinityV308FemurExpertDeployedVal__.../fold_0/

├── stage1_router/stage1_target_router_fold0.joblib

└── split_candidate_rf/v35_candidate_split_aware_rf_all_candidates.joblib
```

5. Local evaluation

The primary result is nested case-grouped out-of-fold evaluation on the frozen 68-patient / 132-anatomy official-aligned local proxy. It is not a hidden-test leaderboard score.

Decoder	Dice	HD95 mm	ASSD mm	Recall	Precision
v3.5 safe base	0.855649	6.686971	1.852273	0.931412	**0.931197**
v3.6.1 split-aware RF OOF	**0.856133**	**6.439807**	**1.789200**	0.931412	0.928409

There were 112 candidate actions. The nested OOF policy selected two candidates in two cases: one reduced a merge error and neither increased merge errors. The total split increase was four. The deployment regressors are then fit on all 112 candidates; any full-fit replay on these 68 cases is treated only as a runtime sanity check, not as unbiased evidence.

6. Reproducibility and limitations

- Given the frozen local base/proposal arrays, the submission candidate extractor reproduces all 112 training candidate keys and all 44 features exactly (maximum absolute error 0.0).
- Candidate fitting and policy selection use case-grouped folds; no case is shared between an OOF training and test fold.
- A non-root GPU container smoke test on pelvic case 039 applied one live LeftHip candidate. Against a separate live v3.5-safe-base run it changed Merge by -1, Split by +2, Dice by +0.0703, HD95 by -2.71 mm and ASSD by -0.814 mm. Femur case 268 generated one candidate and conservatively rejected it. Offline cached Stage-A crops are not byte-identical to live container crops, so these two live checks complement rather than replace nested OOF.
- The sample of candidate actions is small. Hidden-test generalization and the 10-minute platform runtime must be checked before activation.
- The RF does not create foreground voxels or remove predicted bone. It only relabels a selected piece of one base instance with a free anatomy-local ID.

7. Runtime and licensing

The image targets an NVIDIA T4 16 GiB GPU. Stage A and each Stage-B expert are loaded serially; the small CPU RF bundle is loaded only after GPU Stage-B inference. STU-Net-B and nnU-Net are used under their respective open-source licenses; Stage-B initialization derives from the official TotalSegmentator `base_ep4k` checkpoint.

8. References

- Isensee, F. et al. nnU-Net, *Nature Methods* 18, 203-211 (2021).
- STU-Net / TotalSegmentator bone pretraining.
- PENGWIN 2026 Task 1 baseline: `github.com/YzzLiu/PENGWIN2026_Task1_AutoSeg_Baseline`.